c) $D(x,y,z)$: z is the disjunction of x and y.

$\exists x\, (Co(x) \wedge \exists y\, Co(y) \wedge D(x,y,z) \wedge T(z))$

d) $C(x,y,z)$: z is the conjunction of x and y

$\forall x \forall y\, (T(x) \wedge T(y) \wedge C(x,y,z) \Rightarrow T(z))$

31) Suppose that the domain of $Q(x,y,z)$, consists of triples x, y, z, where $x = 0, 1$ or 2, $y = 0$ or 1, and $z = 0$ or 1. Write out these propositions using disjunctions and conjunctions.

a) $\forall y\, Q(0,y,0)$: $Q(0,0,0) \wedge Q(0,1,0)$

b) $\exists x\, Q(x,1,1)$: $Q(0,1,1) \vee Q(1,1,1) \vee Q(2,1,1)$

c) $\exists z\, \neg Q(0,0,z)$: $\neg Q(0,0,0) \vee \neg Q(0,0,1)$

d) $\exists x\, \neg Q(x,0,1)$: $\neg Q(0,0,1) \vee \neg Q(1,0,1) \vee \neg Q(2,0,1)$

33) Express each of these statements using quantifiers. Then form the negation of the statement, so that no negation is to the left of a quantifier. Next express the negation in simple English. (Do not simply use the phrase "It is not the case that.")

a) Some old dogs can learn new tricks.

Domain (the set of dogs). $O(x)$: x is old $T(x)$: x can learn new tricks.

$\exists x\, (O(x) \wedge T(x))$.

Negation: $\neg \exists x (O(x) \wedge T(x)) \equiv \forall x \neg (O(x) \wedge T(x))$ (De Morgan's law of quantifiers)

$\equiv \forall x\, (\neg O(x) \vee \neg T(x))$ (1st De Morgan's Law)

Either all dogs are not old or they cannot learn new tricks.

b) No rabbit knows calculus.

Domain: The set of animals $R(x)$: x is a Rabbit

$C(x)$: x knows Calculus.

$\forall x\, (R(x) \Rightarrow \neg C(x))$ Negation: $\neg \forall x (R(x) \Rightarrow \neg C(x))$

$\equiv \exists x \neg (R(x) \Rightarrow \neg C(x))$ (De Morgan's law of quantifiers)

$\equiv \exists x\, \neg (\neg R(x) \vee \neg C(x))$ (Disjunction conditional equivalence)

$\equiv \exists x\, \neg (\neg (R(x) \wedge C(x)))$ (1st De Morgan's Law)

$\equiv \exists x\, (R(x) \wedge C(x))$ (Double negation law)